INTERNSHIP REPORT

ON

Automation of Coursera Web Application To Identify Courses

Yeshwantrao Chavan College of Engineering

Bachelor of Technology in

Computer Science and Engineering

Submitted by:

Vaishnavi Subhash Asare

Supervised by:

Prof. Chanchla Tripathi
(Faculty Supervised)

Ashly M. Benny
(Industry Supervisor)

Nagar Yuwak Shikshan Sanstha's

YESHWANTRAO CHAVAN COLLEGE OF ENGINEERING,

(An Automation Institution Affiliated to Rashtrasant Tukadoji Maharaj Nagpur University, Nagpur)

NAGPUR – 441 110

2025-26

CERTIFICATE OF APPROVAL

Certified that the Internship project entitled **“Automation of Coursera Web Application to Identify Courses”** has been successfully completed by **Vaishnavi Subhash Asare** during session 2025-26.

Faculty Supervisor

Signature :

Name : Prof. Chanchla Tripathi

Designation : Assistant Professor

Industry Supervisor

Signature :

Name : Ashly M. Benny

Designation : Contractor, CPO L&D GenC Training

ACKNOWLEDGEMENT

I, **Vaishnavi Subhash Asare** would like to convey my gratitude to Yeshwantrao Chavan College of Engineering **Ashly M. Benny, Contractor CPO L&D GenC Training, Cognizant Technology Solution** for emphasizing on the Semester Internship Program and giving me the platform to interact with industry professionals.

I would also like to thank **Prof. Chanchla Tripathi** and **Dr. Lalit Damahe** for giving me the opportunity to work on the prestigious Internship

I extend my warm gratitude and regards to everyone who helped me during my internship.

13-Jan-2026

Candidate ID: 39901857

Vaishnavi Subhash Asare
B.Tech Communication & Computer Science
Yeshwantrao Chavan College of Engineering, Nagpur

Dear **Vaishnavi Subhash Asare,**

Further to our Letter of Intent for the position of Programmer Analyst Trainee / Programmer Analyst aligned to the hiring category, we are pleased to offer you an internship with us at Cognizant office for a **period of 3 to 6 months**. Your internship on-boarding will be scheduled based on your availability, factoring your college exam schedule, availability of your Provisional Certificate and our business requirements.

During this period, you will be provided with a stipend of **INR 12,000**/- equated to the planned duration of the Internship curriculum and will be paid monthly based on your attendance, subject to eligibility for the monthly payroll processing for a given month, except for the last month's stipend, which will be paid upon successful completion of the learning curriculum.

Though Cognizant Internship is a skilling program aimed at enhancing technical acumen, it does not guarantee employment and there is no employer – employee relationship during the course of this internship program. However, the successful completion of internship may form a critical part of your eligibility for employment with Cognizant if an opportunity arises in future.

You will be provided a learning curriculum as per the skill track assigned to you. The learning design would expect you to drive your learning through hands on exercise and project work. There will also be a series of webinars, quizzes, Subject Matter Expertise (SME) interactions, mentor connects, code challenges, assessments etc. to accelerate your learning. The outcome during Internship would be monitored through formal evaluations.

Prior to joining on the rolls of Cognizant, if offered, you must have successfully completed the prescribed Internship program. In the event of unsatisfactory performance during the Internship or non-completion of the Internship, no Internship Completion Certificate shall be issued by Cognizant. Cognizant reserves rights at its sole discretion to revoke its Letter of Intent.

Section A: Terms and Conditions:

1. The Internship timings would be for 10 hours per day from Monday through Friday, aligned with the working timings followed in Cognizant, which, based on the need, could also be operated on a shift model. Attendance is mandatory on all the days to stay active in the Internship Program. The Internship Offer would be cancelled if the mandatory requirement of minimum 85% attendance at office is not met in a month.

2. Interns are covered under Cognizant's calendar holidays of the respective location of internship, and you would need to adhere with minimum attendance requirements. Prior approvals are must towards any unavoidable leave or break requests during the program and the internship would be cancelled if leaves are availed without prior approvals.

3. You would be required to ensure timely completion and submission of assignments, project work and preparation required prior to the sessions, failing which your internship would be cancelled.

4. The Technical skills track mapped could change at the start or mid-way or even later during the program, depending on business demand changes and you would be required to be flexible for this change, failing which your internship would be cancelled.

5. Stipend payment will be done for the prescribed Internship Curriculum period only and no additional payment will be done for any delay in completion. Minimum 85% of attendance is mandatory for the successful processing of your monthly stipend payment.

6. There would be zero tolerance to plagiarisms and misconduct during the internship. Adherence to Cognizant Internship policies and guidelines is mandatory and any breach or incident reported will lead to immediate cancellation of Internship without any notice. You would be required to complete Cognizant mandatory trainings such as Code of Conduct and Acceptable Use Policy (AUP) within the given timelines.

7. During the course of your Internship and at all times, you shall be governed by Cognizant's Social Media Policy and shall, refrain from posting malicious, libelous, defamatory, false, obscene, political, anti-social, abusive, and threatening messages/statements or disparaging the Company, clients, associates, competitors, or suppliers or any third parties, irrespective of whether any such statements are likely to cause damage to any such entity or person. Any breach of this section would lead to immediate cancellation of the Internship.

8. Cognizant reserves rights regarding IT infra as applicable and access to information and materials of Cognizant during the internship period and may modify or amend the Cognizant GenC program terms and conditions from time to time.

9. It is hereby clarified that participation in this Internship shall not constitute you to be an employee of Cognizant nor shall it obligate Cognizant for any purpose whatsoever. The scope of this Internship does not include any supervisory responsibilities and that there is no agency, fiduciary or employer-employee relationship intended or created by reason of this document.

10. Cognizant holds all rights to cancel this Internship Offer owing to any reason at any point in time during your Internship, including due to non-conformance of performance benchmark or moral code of conduct or in case of you failing to participate in the Internship within the given date/timeline or for any other reasons upon providing written communication of the same to you. Upon such cancellation of this Internship Offer, your access and participation in the Internship shall stand cancelled.

11. At the time of your reporting for the internship, you will be required to sign a Non – Disclosure Agreement with the Company. During the course of your internship and after completion of the same, you are required to maintain strictest confidentiality with respect to Company proprietary or products that you access or come into contact with, during your project as an Intern, at all times as per our Policy. Use of company proprietary information or products shall not be made without prior permission from the concerned authority. Any breach of information security will be dealt as per Company Policy.

12. After successful completion of your internship if there is a business demand which expects you to get enabled on a different skill, you would be onboarded as full-time employee (FTE) and deployed into another formal training based on business demand to a specific skill track which will be used as basis towards your allocation to projects/roles. Successful completion of this training is mandatory to continue as an FTE with Cognizant.

13. This offer from Cognizant shall be active and **valid for only 3 calendar days** and hence you are expected to accept or decline the offer through the Company's online portal within the said time-period of 3 calendar days and you will also be required to submit the mandatory documents at least 10 **days** before your Internship Onboarding Date as part of your Pre-joining & Background Verification (BGV) process. In case you do not comply to the above timelines or if the background verification checks reveal unfavorable results at any time, this Offer shall stand withdrawn and will be considered as cancelled. Any official written extension to the offer validity and the above-mentioned timelines will be at the sole discretion of Cognizant.

14. For avoidance of doubt, it is herewith stated that the Internship shall stand cancelled on the below scenarios as well:

a. In the event of you accepting this Internship Offer but not joining into the Internship on the specified date

and at the specified location of on-boarding.
b. In the event of you not accepting this Internship Offer or failing to communicate acceptance within 3 calendar days as stated above, the opportunity to do internship will be cancelled.
c. For such other operational, regulatory reasons including breach of terms herein.

Thereupon, your access shall also stand revoked, and Cognizant shall not be obligated to extend nor be liable for any claims due to cancellation of this Internship Offer.

On the occurrence of any of the above-mentioned scenarios **(Refer to Section A: Terms and Conditions)**, Cognizant reserves the right to revoke your Internship Offer at its sole discretion. Upon cancellation of your Internship Offer your Letter of Intent would also be revoked at the sole discretion of Cognizant.

Below are the **mandatory documents** to be submitted as part of your **Background Verification:**
• Your Pan Card

Below are the **mandatory documents** to be submitted as part of your **Pre- joining formalities:**
• 2 Passport sized Photographs preferably with a White background
• Personal individual bank account from a nationalized bank for processing stipend

In case of additional queries or concerns, you can raise a query at
https://campus2cognizant.cognizant.com

We wish you good luck.

Yours sincerely,
For Cognizant Technology Solutions India Pvt. Ltd.,

Atul Sahgal
Senior Vice President - Talent Acquisition

I have read the offer, understood and accept the above mentioned terms and conditions.

Signature: **Date:**

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1. INTRODUCTION OF THE REPORT

1.1 Background

The rapid advancement of digital technology over the past decade has fundamentally transformed how education and professional training are delivered on a global scale. Traditional classroom-based learning has progressively given way to online platforms that offer flexibility, scalability, and accessibility to learners across geographical and socioeconomic boundaries. Among these platforms, Coursera has emerged as one of the most widely used and reputable online learning ecosystems, offering thousands of courses spanning diverse domains such as programming, data science, artificial intelligence, language learning, business management, and more. With millions of active users worldwide, maintaining the reliability, performance, and seamless user experience of such platforms is not merely desirable — it is a critical operational requirement.

In the contemporary software development landscape, Quality Assurance (QA) has evolved from being a post-development activity to an integral part of the software delivery lifecycle. QA professionals are responsible for validating that applications meet functional requirements, perform efficiently under varied conditions, and deliver a consistent experience to end users. Within QA, web automation testing has gained significant prominence as a methodology for simulating real user interactions programmatically. Unlike manual testing, which is inherently time-consuming, repetitive, and susceptible to human error, automation testing enables teams to execute large suites of test scenarios consistently, rapidly, and with greater precision. It also enables continuous integration practices where tests are automatically triggered as part of the development pipeline.

This internship was undertaken at Cognizant Technology Solutions, a globally recognized Information Technology and consulting firm known for its expertise in digital transformation, software testing, and enterprise solutions. During the internship, the student was introduced to industry-standard automation testing practices and was assigned a project focused on automating critical user workflows on the Coursera web platform. Using browser automation tools, the student designed and implemented scripts that simulate real user interactions, extract course-related data, and validate form functionality. This hands-on exposure provided an invaluable bridge between academic learning and professional practice, equipping the student with relevant, market-ready technical skills.

1.2 Purpose of the Report

This report has been prepared to fulfil the requirements of the Semester Internship Program as prescribed by Yeshwantrao Chavan College of Engineering, Nagpur, under the Bachelor of Technology in Computer Technology program for the academic year 2025–26. It serves as a formal documentation of the internship experience, the technical work undertaken, and the professional competencies developed during the tenure.

The report serves the following specific purposes:

- To provide a comprehensive and structured account of the internship experience undertaken at Cognizant Technology Solutions, under the guidance of the designated industry mentor and faculty supervisor.
- To document the problem statement, objectives, methodology, tasks performed, observations, and outcomes of the automation project carried out on the Coursera web application.
- To demonstrate the practical application of technical knowledge and skills acquired during the B.Tech program in a real-world, professional industry environment.
- To reflect upon the learning outcomes, challenges encountered, and competencies developed during the internship period, contributing to the student's holistic professional growth.
- To serve as a reference document for future students and faculty, illustrating how industry projects align with academic curricula in the domain of software testing and web automation.

1.3 Objective of the Internship

The following objectives were defined in consultation with the industry mentor at the commencement of the internship. These objectives guided the planning, execution, and evaluation of the project throughout its duration:

- To understand and apply web automation testing concepts in a real-world industry setting under professional supervision, thereby bridging theoretical knowledge with practical implementation.

To design and implement automation scripts capable of identifying and extracting course-related data, including course names, learning hours, and ratings, from the live Coursera platform.

- To handle dynamic web elements including search bars, dropdown menus, multi-window navigation, and interactive web forms using industry-standard browser automation techniques.

- To perform form validation testing by deliberately submitting incorrect or improperly formatted input data and programmatically capturing the resulting error messages generated by the application.
- To develop proficiency in industry-standard automation tools and frameworks — including Selenium WebDriver, TestNG, and ExtentReports — and to apply them effectively to solve practical QA challenges.
- To enhance key professional skills including technical documentation, structured problem-solving, analytical thinking, and collaborative teamwork within a professional environment.
- To gain exposure to Behaviour Driven Development (BDD) practices and understand how Gherkin syntax facilitates communication between technical and non-technical stakeholders in agile software teams.

1.2 Scope of Work

The internship project was scoped around three specific and well-defined automation tasks to be performed on the Coursera website (coursera.org). Each task was designed to address a distinct aspect of web automation — data extraction, category navigation, and form validation — collectively covering a broad spectrum of real-world QA scenarios. The scope was deliberately focused to ensure depth of implementation and quality of output within the available internship duration.

Task 1 — Web Development Course Identification: To search for beginner-level web development courses offered in the English language on the Coursera platform, and to extract and programmatically display the course name, total estimated learning hours, and learner rating for the first two courses appearing in the search results. This task involved handling Coursera's dynamic search interface, applying filters, and accurately reading structured course data from the results page.

Task 2 — Language Learning Data Extraction: To navigate to the Language Learning category on Coursera and systematically extract all available languages along with their corresponding proficiency levels. The task further required the script to display the total number of courses available under each language and proficiency level combination, thereby producing a structured summary of Coursera's language learning catalogue.

Task 3 — Form Validation and Error Capture: To navigate from the Coursera homepage to the 'For Enterprise' section, access the 'Courses for Campus' product page under the Products menu, and interact with the 'Ready to Transform' contact form. The task involved filling the form with one intentionally invalid input — specifically an incorrectly formatted email address — and programmatically capturing and displaying the validation error message triggered by the form's built-in input validation logic.

1.3 Tools and Technologies Used

Tool/Technology	**Purpose**
Selenium WebDriver	Browser automation and web element interaction
BDD(Behaviour Driven Development)	Writing test scenarios in plain English using Gherkin syntax (Given/When/Then)
TestNG	Test execution framework for managing and running automated test cases
ExtentReports	Generating detailed HTML test execution reports with logs and results
GitHub	Version control and source code management
Java	Primary programming language for automation scripting
ChromeDriver	WebDriver for Google Chrome browser
Coursera.org	Target web application for testing
IntelliJ IDE	Development and script execution environment

1.4 Methodology

The internship project was executed following a structured and systematic approach, aligned with standard software development and QA practices. The workflow was divided into five clearly defined phases, each building upon the outcomes of the previous one, to ensure organized progression from problem understanding to final documentation.

Phase 1 — Requirement Analysis: The problem statement provided by the industry mentor was carefully analyzed to identify all relevant web elements, user interaction flows, and expected outputs for each of the three automation tasks. This phase involved a detailed study of the Coursera website structure, including its navigation patterns, dynamic elements, and form behaviors, which informed the design decisions made during script development.

Phase 2 — Environment Setup: All required software tools, including the Java Development Kit (JDK), IntelliJ IDEA, Selenium WebDriver, ChromeDriver, TestNG, and ExtentReports libraries, were installed and configured on the development system. Project dependencies were managed using Maven, and the workspace was initialized with the appropriate directory structure to support BDD-based test development.

Phase 3 — Script Development: Automation scripts were written in Java to perform each of the three defined tasks. This phase involved identifying web elements using appropriate locator strategies (such as CSS selectors, XPath, and element IDs), implementing logic for dynamic content handling, applying explicit and implicit waits to manage page load behavior, and writing assertions to verify expected outcomes. BDD feature files were created using Gherkin syntax, with corresponding step definition classes implemented in Java.

Phase 4 — Execution and Observation: The developed scripts were executed against the live Coursera website. The outputs were carefully observed, verified against expected results, and recorded. Any discrepancies or execution failures encountered during testing — such as element loading delays or platform-side changes — were analyzed and resolved through code adjustments. ExtentReports was used to generate detailed HTML execution reports for each test run.

Phase 5 — Documentation: All findings, observations, code samples, test results, and outcomes were systematically compiled and documented into this internship report, following the format and guidelines prescribed by Yeshwantrao Chavan College of Engineering, Nagpur. The report reflects the complete journey from project initiation to successful completion of the defined automation objectives

2. OVERVIEW OF THE INTERNSHIP PROVIDING ORGANIZATION

2.1 About Cognizant Technology Solutions

Cognizant Technology Solutions is one of the world's leading professional services companies, dedicated to helping clients transform their business, operating, and technology models for the digital era. Headquartered in Teaneck, New Jersey, USA, Cognizant operates across multiple continents and serves a diverse clientele spanning industries such as banking and financial services, healthcare, retail, manufacturing, communications, and technology. With a global workforce of over 300,000 skilled professionals, the company combines a deep passion for client satisfaction, continuous technology innovation, industry-specific domain expertise, and a community-minded approach to drive meaningful and sustained digital transformation at scale.

Founded in 1994 as an in-house technology unit of Dun and Bradstreet, Cognizant has grown over three decades into a Fortune 500 company and a globally recognized leader in IT services. The company is listed on the NASDAQ stock exchange under the ticker symbol CTSH and is consistently ranked among the most admired and trusted organizations in the technology services sector. Its remarkable growth trajectory reflects not only its ability to adapt to a rapidly changing technology landscape but also its unwavering commitment to delivering measurable value to its clients, employees, and stakeholders across the world.

Cognizant has earned widespread recognition for fostering a strong culture of continuous learning, investing significantly in talent development programs, and maintaining a relentless focus on quality and innovation. Its service portfolio spans digital engineering, cloud infrastructure, artificial intelligence and analytics, enterprise application development, and quality assurance — making it a comprehensive and trusted technology partner for organizations at every stage of their digital transformation journey. The company operates through an extensive network of delivery centers, innovation hubs, and regional offices across North America, Europe, Asia-Pacific, and other regions, enabling seamless global service delivery and round-the-clock client support.

2.2 Quality Engineering and Assurance (QEA) – Service Line

The internship was undertaken under the Quality Engineering and Assurance (QEA) service line of Cognizant, which is a dedicated and strategically important division focused on ensuring the quality, reliability, performance, and security of enterprise applications, business processes, and complex technology systems. As one of Cognizant's most critical service lines, QEA plays a central and proactive role in helping clients achieve the highest standards of software quality across the entire development and delivery lifecycle.

Cognizant's QEA service line offers a comprehensive and continuously evolving portfolio of services, including intelligent and automated quality assurance, modernization assurance, and experience assurance. These services are carefully designed to accelerate business and technology changes, improve end-user experiences, minimize operational risk, and ensure regulatory compliance across a wide range of industry domains. The QEA division partners with clients from the initial stages of requirements gathering through to post-deployment validation, embedding quality practices at every phase of the software delivery process rather than treating quality as an afterthought.

A defining characteristic of the QEA division is its strong emphasis on automation and Artificial Intelligence as key enablers of testing efficiency and quality excellence. By leveraging AI-driven testing tools, robotic process automation, intelligent test generation, and advanced analytics, the QEA team is able to significantly reduce testing cycle times, improve defect detection rates, and deliver superior outcomes with reduced manual effort. This forward-looking, technology-first approach allows Cognizant's clients to achieve digital success while maintaining a strong competitive advantage in an increasingly complex and fast-evolving digital landscape. The division has successfully partnered with major global enterprises across healthcare, finance, retail, and manufacturing sectors to fundamentally transform their quality assurance processes and achieve measurable improvements in software reliability, regulatory compliance, and overall time-to-market performance.

2.3 Generation Cognizant (GenC) – Internship Program

The internship was part of Cognizant's Generation Cognizant (GenC) program, a structured, comprehensive, and highly regarded learning initiative designed specifically to engage young engineering talent from institutions across India. The GenC program provides interns and fresh graduates with a clearly defined technical learning pathway that systematically equips them with the skills, professional mindset, and industry readiness required to contribute meaningfully to real-world software engineering projects from the very beginning of their careers.

Through the GenC program, participants are given the opportunity to interact closely with experienced Subject Matter Experts (SMEs), gain a thorough and practical understanding of the corporate work environment, and develop both technical and interpersonal competencies necessary to function effectively as software engineers in a fast-paced, delivery-oriented industry setting. Rather than adopting a passive, lecture-based learning model, the GenC program is built on the foundational principle of Learner Autonomy, wherein students are encouraged and empowered to take full ownership of their learning journey using a variety of curated tools, resources, and self-paced digital platforms. The emphasis throughout the program is consistently on learning by doing — empowering interns to build genuine problem-solving capabilities through hands-on coding practice, guided project execution, and exposure to real-world industry challenge scenarios.

The program adopted the Flipped Classroom methodology as its core pedagogical model, effectively combining self-paced independent study with expert-guided interactive practice sessions. During designated self-learning hours, interns studied course content, watched instructional videos, and completed structured exercises independently through the Udemy online learning platform and the Tekstac coding environment — a specialized platform designed to support hands-on coding practice with automated evaluation and performance tracking. These self-directed learning sessions were complemented by live practice sessions facilitated by experienced Cognizant SMEs, who guided discussions, addressed conceptual doubts, demonstrated practical real-world applications, and actively mentored interns in applying their newly acquired knowledge to solve realistic problem statements modeled on genuine industry scenarios.

2.4 Program Structure and Learning Journey

The GenC program followed a meticulously structured 65-day learning journey, thoughtfully divided into two progressive and interconnected stages that built systematically upon one another to deliver a holistic, industry-aligned, and practically relevant skill development experience for all participants.

Stage 1 — Core Java with SQL formed the essential foundational phase of the program. This stage provided comprehensive coverage of fundamental programming concepts, including Core Java basics, Object-Oriented Programming (OOP) principles such as inheritance, encapsulation, abstraction, and polymorphism, the Java Collections framework, Exception Handling mechanisms, File Handling operations, Java Database Connectivity (JDBC), and Structured Query Language (SQL) for relational database interaction. Mastery of these topics was a critical prerequisite, as they collectively provided the technical foundation and programming discipline necessary to write efficient, maintainable, and scalable automation code in the stage that followed. Interns were required to complete hands-on coding exercises, pass module assessments, and successfully demonstrate understanding through evaluated practical tasks before progressing to the next stage.

Stage 2 — Functional Testing with Selenium Automation Concepts constituted the advanced and specialized phase of the program, covering the full professional spectrum of web automation testing in a structured and progressively challenging manner. This stage began with the core principles of Functional Testing and systematically introduced Selenium WebDriver for browser-based automation, the Page Object Model (POM) design pattern for creating maintainable test frameworks, Apache POI for enabling data-driven testing using Excel files, the TestNG testing framework for test management and execution, BDD Cucumber for behaviour-driven development using Gherkin syntax, and a suite of DevOps tools including Git for version control, Maven for build and dependency management, Jenkins for continuous integration pipeline setup, and Selenium Grid for distributed parallel test execution. The stage additionally covered API automation techniques using Postman and REST Assured, as well as an introductory module on Generative AI concepts and their emerging practical applications in modern software testing workflows.

The program culminated in a Business Aligned Project phase, structured into two components: a Mini Project undertaken individually and a Hackathon executed as a collaborative team activity. Both components were carefully designed to simulate authentic real-world project delivery environments. They were formally evaluated by Technical Subject Matter Experts from the Business Unit through a structured two-stage assessment process comprising an Interim Evaluation and a Final Evaluation, both conducted via video interviews. This evaluation methodology ensured that interns demonstrated not only strong technical competency but also the professional communication and presentation skills essential in an industry setting.

2.5 Internship Environment and Supervision

The internship was conducted in a well-structured, professionally supervised environment that closely replicated the working conditions and collaborative dynamics of a real software engineering team at Cognizant. The daily schedule comprised a carefully balanced combination of self-paced learning modules, hands-on coding exercises, live code reviews, project deliverable submissions, and collaborative group activities — all thoughtfully orchestrated to maximize learning outcomes, maintain consistent engagement, and build professional discipline within the defined program duration.

Industry supervision was provided by Ashly M. Benny, Contractor, CPO Learning and Development, GenC Training, Cognizant Technology Solutions, who served as the designated Industry Supervisor throughout the internship. Under her expert guidance, the student received consistent technical mentorship, clear project direction, structured feedback on deliverables, and ongoing performance evaluations, ensuring that all work undertaken was firmly aligned with industry standards, professional expectations, and best practices in software quality assurance.

Academic supervision was concurrently and actively provided by Dr. Piyush Ingole, Assistant Professor, Department of Computer Technology, Yeshwantrao Chavan College of Engineering, Nagpur, who served as the Faculty Supervisor. His guidance ensured that the internship experience was meaningfully connected to the academic objectives of the B.Tech program and that all documentation, reporting, and project outputs met the prescribed institutional standards and evaluation criteria.

Regular SME-connect sessions, one-on-one mentor interaction meetings, and structured group review discussions were integral and recurring features of the internship experience. These sessions fostered a highly collaborative and intellectually stimulating learning environment in which interns could seek expert guidance, share progress updates, discuss technical challenges openly, and receive constructive and actionable feedback. This dual-supervision model — seamlessly combining industry expertise with dedicated academic oversight — ensured that the student's overall internship experience was both technically rigorous and academically enriching, resulting in a truly well-rounded and impactful professional development outcome.

3. ABOUT THE PROJECT

3.1 Project Title

Automation of Coursera Web Application to Identify Courses

3.2 Project Overview

This project was undertaken as part of the Hackathon component of the Cognizant GenC Quality Engineering and Assurance (QEA) internship program. The project involved the design and implementation of a comprehensive web automation testing framework targeting the Coursera platform (coursera.org), one of the world's most widely used online learning portals.

The primary objective of the project was to automate three distinct real-world user workflows on the Coursera website, covering a broad range of automation scenarios including browser-based search and data extraction, dynamic dropdown handling, multi-window navigation, web form interaction, and validation error capture. The project was built using Java as the programming language, Selenium WebDriver for browser automation, BDD Cucumber for behaviour-driven test development, TestNG as the test execution framework, ExtentReports for test reporting, and GitHub for version control. The overall framework was designed following the Page Object Model (POM) design pattern to ensure maintainability, reusability, and scalability of the automation codebase.

3.3 Problem Statement

The project addressed the following problem statement as defined in the Hackathon:

"Search and display all web development courses. The courses should be at beginner level, offered in the English language. Display the first two courses with course name, total learning hours, and rating."

The broader scope of the Hackathon extended this problem statement to include three specific automation tasks:

Task 1: Search for web development courses for beginners level in the English language on Coursera and extract the course name, total learning hours, and rating for the first two courses displayed in the search results.

Task 2: Navigate to the Language Learning category on Coursera, extract all available languages and their different proficiency levels along with the total course count for each, and display the complete extracted data.

Task 3: From the Coursera homepage, navigate to "For Enterprise," access "Courses for Campus" under the Products section, fill in the "Ready to Transform" contact form with one intentionally invalid input such as an incorrectly formatted email address, capture the error or warning message displayed by the form validation, and display it.

3.4 Key Automation Scope

The automation framework built for this project covered the following key technical scenarios:

Handling different browser windows and search functionality was a core requirement, as the Coursera platform dynamically loads content and opens results in the same or new browser contexts depending on the user action. The framework was designed to manage these scenarios using Selenium WebDriver's window handling capabilities.

Extracting multiple dropdown list items and storing them in collections was addressed in Task 2, where the Language Learning section presented multiple dropdown-style filter menus containing language names and proficiency levels. These were extracted programmatically and stored in Java collections such as ArrayLists and HashMaps for structured display and reporting.

Navigating back to the homepage was required between tasks to simulate realistic user journeys. The framework incorporated explicit navigation logic to return to the Coursera homepage using both browser history navigation and direct URL loading.

Filling forms across different web page objects was required in Task 3, where the "Ready to Transform" form contained multiple input fields of different types including text fields, dropdown selectors, and submit buttons, all located within a complex multi-section page layout.

Capturing warning messages was the final objective of Task 3, where the form validation error displayed upon submitting an invalid email address was captured programmatically using Selenium element identification and stored for reporting.

Scrolling down in web pages was necessary throughout the project, as several key elements on the Coursera platform were not visible within the initial viewport and required JavaScript-based or Actions class-based scrolling to bring them into view before interaction.

3.5 Tools and Technologies Used

Tool/Technology	**Purpose**
Selenium WebDriver	Browser automation and web element interaction
BDD(Behavior Driven Development)	Writing test scenarios in plain English using Gherkin syntax (Given/When/Then)
TestNG	Test execution framework for managing and running automated test cases
ExtentReports	Generating detailed HTML test execution reports with logs and results
GitHub	Version control and source code management
Java	Primary programming language for automation scripting
ChromeDriver	WebDriver for Google Chrome browser
Coursera.org	Target web application for testing
IntelliJ IDE	Development and script execution environment
Apache POI 5.2.4	Reading and writing test data from Excel files for data-driven testing
JIRA	Agile project management, test case documentation, sprint management, and defect tracking

Maven	Build automation and dependency management
Page Object Model (POM)	Design pattern for separating test logic from page element definitions

3.6 Framework Architecture

The automation framework built for this project covered the following key technical scenarios:

Handling different browser windows and search functionality was a core requirement, as the Coursera platform dynamically loads content and opens results in the same or new browser contexts depending on the user action. The framework was designed to manage these scenarios using Selenium WebDriver's window handling capabilities.

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Scrolling down in web pages was necessary throughout the project, as several key elements on the Coursera platform were not visible within the initial viewport and required JavaScript-based or Actions class-based scrolling to bring them into view before interaction.

3.7 Task 1 – Web Development Course Identification

Objective: To search for beginner-level web development courses offered in English on Coursera and extract the name, total learning hours, and rating for the first two courses in the results.

Steps Performed:

The browser was launched and maximized, and the Coursera homepage at coursera.org was loaded. The search bar was located using its XPath locator and the keyword "Web Development" was entered and submitted. On the search results page, the Level filter was applied by selecting "Beginner" from the filter panel on the left side, and the Language filter was set to "English." The page was scrolled down to ensure all course cards were loaded within the viewport. The first two course cards were identified using dynamic XPath locators targeting the course card container elements. For each of the two courses, the course name, total learning hours, and rating were extracted using the getText() method of Selenium WebDriver and stored in a Java ArrayList. The extracted data was then printed to the console and logged into the ExtentReport for documentation.

Challenges Faced: The search results page used dynamic lazy loading, meaning course cards only fully rendered after the page was scrolled. Handling this required the use of JavascriptExecutor for scrolling combined with Explicit Waits using ExpectedConditions to ensure elements were fully visible before data extraction was attempted.

3.8 Task 2 – Language Learning Data Extraction

Objective: To navigate to the Language Learning category on Coursera, extract all available languages and their proficiency levels, and display the total course count for each combination.

Steps Performed:

From the Coursera homepage, the subject catalog was accessed and the Language Learning category was selected. The page loaded a list of language options displayed either as filter dropdowns or category links. Using Selenium's findElements() method with a CSS selector targeting all language filter items, the complete list of available languages was extracted and stored in a Java ArrayList. For each language, the proficiency level options such as Beginner, Intermediate, Advanced, and Mixed were identified from the corresponding dropdown or sub-filter panel. The total number of courses available for each language-level combination was extracted from the count label displayed alongside each filter option. All extracted data was stored in a HashMap with the language name as the key and a nested structure of levels and counts as the value, and was then displayed in a formatted output and logged to the ExtentReport.

Challenges Faced: The filter panel containing language and level options was dynamically rendered and required scrolling within the filter container div. Standard scrolling using JavascriptExecutor was used to bring nested filter items into view. Storing multi-level nested data required careful use of Java collections including HashMap and ArrayList to preserve the hierarchical structure of the extracted information.

3.9 Task 3 – Form Validation and Error Message Capture

Objective: To navigate from the Coursera homepage to the "For Enterprise" section, access the "Courses for Campus" product page, fill the "Ready to Transform" contact form with an invalid email input, and capture the resulting error message.

Steps Performed:

The automation script started from the Coursera homepage. The "For Enterprise" option was located in the top navigation bar using a link text locator and clicked. On the For Enterprise landing page, the Products menu was accessed and the "Courses for Campus" option was selected. The script scrolled down the Courses for Campus page until the "Ready to Transform" contact form was visible within the viewport using JavascriptExecutor. The form fields were located using their XPath or CSS ID locators. All mandatory fields such as First Name, Last Name, and Organization were filled with valid test data. The email field was intentionally populated with an incorrectly formatted email address such as "testuser@" to trigger the form's built-in validation. The submit button was clicked, and the script waited for the validation error message to appear using an Explicit Wait. The error message text was captured using the getText() method and stored in a String variable, then printed to the console and logged in the ExtentReport.

Challenges Faced: The "Ready to Transform" form was positioned well below the page fold and required reliable scrolling logic using JavascriptExecutor to bring it into view. The form contained multiple field types including text inputs and dropdown selectors, each requiring different Selenium interaction methods. The error message was rendered as a dynamic DOM element that appeared only after the submit button was clicked, requiring an Explicit Wait with ExpectedConditions.visibilityOfElementLocated to ensure it was fully rendered before the getText() call was executed.

3.10 Test Execution and Reporting

All three automation tasks were executed as part of a unified TestNG test suite configured through a testng.xml file. The tests were grouped into a regression suite and executed sequentially. Upon completion of each test run, ExtentReports automatically generated a detailed HTML report containing the test execution summary, individual step results, pass/fail status, and captured screenshots at key checkpoints.

The test suite was also integrated with Jenkins to enable scheduled and on-demand CI execution. The project source code was maintained on GitHub with regular commits following standard Git workflow practices including branching, committing, and pushing changes. JIRA was used throughout the project for tracking user stories, managing sprint backlogs, logging defects, and maintaining test case traceability.

3.11 Defect Management Using JIRA

All defects identified during test execution were logged in JIRA with the following details captured for each defect: a unique defect ID, the defect summary and description, the steps to reproduce the issue, the expected versus actual result, the severity and priority classification, the environment details, and any screenshots or logs attached as evidence. Sprint reviews and retrospectives were conducted as part of the Agile methodology followed during the project, and all defect statuses were tracked through the JIRA board from Open through In Progress to Resolved and Closed.

CONCLUSION

This internship at Cognizant Technology Solutions, as part of the Generation Cognizant (GenC) Quality Engineering and Assurance program, was a highly rewarding experience that successfully bridged the gap between academic knowledge and real-world industry practice. The structured 65-day learning journey covered Core Java, SQL, Functional Testing, Selenium WebDriver, BDD Cucumber, TestNG, ExtentReports, Page Object Model, Apache POI, GitHub, Maven, Jenkins, and API Automation providing a comprehensive foundation in modern software quality assurance practices.

The Hackathon project, centered on automating three user workflows on the Coursera web application, served as the practical culmination of all concepts learned during the program. Through this project, hands-on experience was gained in browser automation, dynamic element handling, dropdown data extraction using Java collections, multi-window navigation, web form interaction, validation error capture, and CI/CD integration using Jenkins and GitHub. The framework was built following industry-standard practices including the Page Object Model design pattern, BDD Cucumber for behaviour-driven test development, and ExtentReports for professional test reporting. JIRA was used throughout for sprint management, test case documentation, and defect tracking in an Agile environment.

Beyond technical skills, the internship developed important professional competencies including structured problem-solving, attention to detail, collaborative teamwork, documentation discipline, and the ability to work within a real corporate environment under the mentorship of experienced industry professionals. Regular SME connect sessions, code reviews, and project evaluations provided continuous feedback that helped refine both the quality of the automation scripts and the overall approach to software testing.

In conclusion, this internship has been a transformative experience that significantly strengthened the technical and professional foundation required for a successful career in software engineering and quality assurance. The skills, tools, frameworks, and industry practices learned and applied during this period will serve as a strong and lasting asset in all future academic and professional endeavors.